

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of M. Sc. Microbiology (Backlog) Examination, September 2019

Course code & Name: MS 861 & Research Methodology

Date: 17/09/2019 Day: Tuesday Time: 10:00 A.M. To 10:20 A.M. Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		10
1.	The following element is an essential component of the “Informed Consent” given by the human subjects participating in clinical research studies.		01
	(i) Voluntarism	(ii) Information disclosed/understood	
	(iii) Consent ability	(iv) All of them	
2.	The following research is mainly concerned with generalizations and with the formulation of a theory.		01
	(i) Fundamental research	(ii) Applied research	
	(iii) Action research	(iv) Exploratory research	
3.	What will be the minimum level of laboratory biosafety containment required in order to do research on Ebola virus?		01
	(i) BSL-1	(ii) BSL-3	
	(iii) BSL-2	(iv) BSL-4	
4.	MeSH terms are preferably used in the _____ section of a biomedical research article for indexing by a bibliographic database.		01
	(i) Material & Methods	(ii) Key Words	
	(iii) Results & Discussion	(iv) References	

5.	Which of the following is a tentative assumption that is made in order to test its logical or empirical consequences in the research process.		01
	(i) Research design	(ii) Research hypothesis	
	(iii) Impact factor	(iv) Research methods	
6.	The following is NOT a plagiarism detection software.		01
	(i) iThenticate	(ii) Turnitin	
	(iii) Plagium	(iv) Scopus	
7.	A type of scientific literature which is not peer reviewed, not indexed by major databases (missing ISBN or ISSN numbers) and is not controlled by commercial publishing is known as		01
	(i) Green literature	(ii) Plagiarized literature	
	(iii) Secondary literature	(iv) Grey literature	
8.	The following is an example of a personal protective equipment (PPE).		01
	(i) N95 respirator	(ii) Gloves	
	(iii) Face shields	(iv) All of them	
9.	The following is an unstructured and a flexible research approach.		01
	(i) Quantitative approach	(ii) Qualitative approach	
	(iii) Pre-determined approach	(iv) All of them	
10.	If a microbiologist conducts experiments on sediment samples of a river estuary to determine the seasonal variation in microbial diversity over several longer periods of time, such type of research can be classified as:		01
	(i) Conceptual research	(ii) Simulation research	
	(iii) Longitudinal research	(iv) One-time research	

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Date: 17/09/2019 Day: Tuesday Time: 10:20 A.M. To 12:00 P.M. Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION – I		
Q - II Answer the following questions as directed (answer any two)		10
1.	What is a research problem? Explain the various considerations to be made for selecting a research problem.	05
2.	Describe the different types of scientific literature.	05
3.	Explain the IMRAD format of a scientific research article.	05
SECTION – II		
Q - III Answer the following questions as directed (answer any five)		15
1.	Define plagiarism. Explain plagiarism of text and self-plagiarism.	03
2.	Explain the criteria of a good research.	03
3.	Describe the features of BSL-3 and BSL-4 laboratories.	03
4.	Explain the “4 R concept” in ethical considerations for research on laboratory animals.	03
5.	Explain the following types of research: (a) Fundamental research (b) Applied research (c) Correlational research	03
6.	What is <i>h</i> -index, how is it determined? State its advantages and disadvantages.	03
7.	Briefly describe the sequential stages of the research process.	03